Lesson 5: Bubble Sort and Efficiency

Module 2 | 45 minutes | Sorting and complexity | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. bubble sort ___ A. Work grows with the square of the input.
b. nested loop ___ B. Smallest-first / biggest-first.
c. ascending / descending ___ C. A loop inside a loop.
d. quadratic ___ D. Repeated passes of neighbour swaps.

2. Plan your code

Before you build it: Bubble sort: swap (i) is defined. Sort scores (5, 3, 8, 1, 9, 2) into ascending order with nested loops, counting comparisons. Write or draw the blocks you will use.

3. Plan your code

Before you build it: Leaderboard order: sort points (40, 90, 10, 70) from biggest to smallest (descending): 90, 70, 40, 10. Write or draw the blocks you will use.

4. Quadratic growth

Bubble sort makes about n × n comparisons. If 10 items take ~100 comparisons, about how many for 100 items? Circle one.

a. A. ~10,000
b. B. ~1,000
c. C. ~200

5. Explain your thinking

Bubble sort is easy but slow for big lists. Explain why, and say where a slow sort might still be fine to use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can implement bubble sort with nested loops.
☐ I can sort ascending or descending.
☐ I can describe how its work grows.